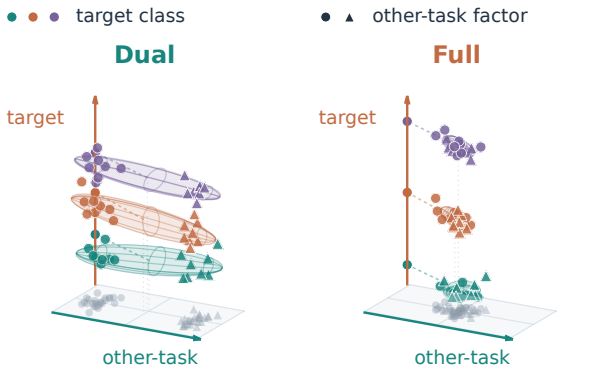
Selective Adaptation under Noisy Supervision: Separating Task Specialization from Representation Transfer

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Abstract

Adapting pretrained models under imperfect supervision raises the problem of reconciling task specialization with the reuse of learned representations. Assessing adaptation therefore requires relating supervision quality and target performance to the reusable information retained. We analyze how selection changes the training measure, how teacher-centered regularization penalizes parameter directions locally, and how the readout shapes the propagation of prediction residuals into feature updates. KPC-CF instantiates these mechanisms with semantic–confidence selection, attention KL, and classifier reset. A gradient decomposition separates support shift from label error, while local analyses characterize anchoring and update pathways. Across two model families, four general classification tasks, and Korean aspect sentiment analysis, these interventions have complementary roles. On Qwen XNLI, Dual outperforms confidence-only selection despite lower selected-label accuracy, gaining 2.97 mean head Macro-F1 points and 0.46 other-task probe points at the longer budget. Full achieves the strongest mean adapted-head performance, exceeding unfiltered adaptation by 5.60 points. Geometry and fresh probes link Full's gain to target specialization; Dual retains stronger cross-task linear information. Attention replay and parameter analysis distinguish structural retention from transferable representations, yielding an endpoint-specific account of how pretrained language models adapt under imperfect supervision.



Δ vs All	All	Conf.	Sem.	Dual	+KL	+Re.	Full
Target head	0.00	-1.38	+2.76	+1.59	+2.81	+4.47	**+5.60**
Other-task probe	0.00	+0.59	+0.46	**+1.05**	+0.91	+0.13	-0.40

Qwen / XNLI · staged 600 · mean, 3 seeds

Figure 1. **Selective adaptation separates task specialization from representation transfer.** Matched schematic points use color for hypothetical target classes and shape for an independent transfer factor. Full illustrates stronger target separation while compressing that factor; Dual retains its variation. Conceptual geometry and arrows accompany the measured footer contrasts. The footer compares all seven Qwen/XNLI variants at staged 600: mean paired Macro-F1 changes from All across three student seeds. Conf./Sem. denote single-score selection; +KL and +Re. denote Dual+KL and Dual+reinit. Row tint scales are separate; Table 1 reports endpoint-specific means and SDs.

1. Introduction

The value of pretrained representations includes their reuse across tasks (Yosinski et al., 2014). For models intended for further reuse, improved target prediction alone does not establish which useful information survives adaptation. Fine-tuning can distort features under distribution shift (Kumar et al., 2022), while translated annotations and model-generated labels can make the evidence driving language adaptation unreliable. The resulting problem is to characterize how learning from this evidence changes task specialization and the structure available for reuse.

Pseudo-labeling and confidence consistency enable learning from unlabeled inputs (Lee, 2013; Sohn et al., 2020). Adaptive thresholds and soft weights address supervision's quantity–quality trade-off (Wang et al., 2023; Chen et al., 2023); SimMatch connects predictions with instance simi-

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larities (Zheng et al., 2022). These signals assess candidate supervision, while selection determines which inputs supply updates. Preserving internal relations and retaining reusable features likewise describe distinct aspects of the adapted representation. This couples support utility, structural retention, and information accessible through different readouts.

Selection changes the measure over training inputs, while a teacher-centered structural penalty supplies local curvature that penalizes parameter motion. The readout shapes a distinct pathway: the map from prediction residuals to feature updates. KPC-CF instantiates these mechanisms with semantic–confidence selection, frozen-teacher attention KL (Wang et al., 2020), and adaptation-onset head reset (Li et al., 2020). Fixed-teacher factorial comparisons across seeds and budgets connect these mechanisms to adapted-head performance, fresh in-task probes, and other-task transfer, with geometry, drift, attention, and parameter analysis.

Qwen XNLI exposes a reversal between supervision precision and adaptation utility. Confidence retains 87.17% correct pseudo-labels versus Dual's 79.96%, yet Dual gains 2.97 mean head Macro-F1 points and 0.46 other-task probe points at matched retention and staged 600 updates across three student seeds. Its 4.70-point gain on the fixed rejected test stratum measures generalization beyond accepted inputs. Full reaches 66.19 head Macro-F1 versus 60.58 for unfiltered adaptation, while Dual leads cross-task transfer (Figure 1). General-task and Korean aspect-sentiment comparisons characterize how these complementary benefits depend on task, backbone, and readout.

Training and evaluation code, fixed configurations, and reproduction instructions are available through our project page and supplementary resources.

2. Related Work

Reliability versus support utility. Confidence selection is affected by calibration and dataset shift (Hendrycks & Gimpel, 2017; Guo et al., 2017; Ovadia et al., 2019). Class balancing (Zou et al., 2018) and curriculum thresholds (Zhang et al., 2021) change the accepted distribution; DeYO tests confidence under destructive transformations (Lee et al., 2024), and FATA augments features to address filtering-induced scarcity (Cho et al., 2025). SimMatch couples semantic and instance consistency through a labeled memory and augmented views (Zheng et al., 2022). Our Qwen selector comparisons instead freeze the teacher and external LaBSE space (Feng et al., 2022), matching retention and update budgets while varying selected identities. This separates label precision from adaptation utility.

Alignment and adaptation assumptions. Adversarial alignment and covariance matching reduce feature discrepancies (Ganin et al., 2016; Sun & Saenko, 2016), but adaptation theory also depends on joint prediction error and label relationships (Ben-David et al., 2010; Zhao et al., 2019). SHOT fixes the source classifier and adapts without source examples (Liang et al., 2020); Ramen selects historical test inputs for online vision-language adaptation (Bao et al., 2026). Our fixed target pool retains labeled source neighbors, allowing support selection to be studied separately from attention anchoring and classifier reset. Distinguishing source–target discrepancy within an encoder from temporal change on identical target inputs prevents these different forms of alignment from being conflated.

Structural retention and feature updates. MiniLM transfers attention distributions and value relations across model sizes (Wang et al., 2020); our final-layer attention KL anchors a student to a frozen teacher of the same architecture. RIFLE uses periodic resets (Li et al., 2020), whereas our adaptation-onset reset isolates its interaction with this anchor. TTE combines averaging, representation diversity, and distillation for online stability (Kim et al., 2025). We connect these mechanisms through the local attention metric and the classifier-dependent encoder gradient, revealing their coupled effects during adaptation.

Readouts and representation evidence. Transfer depends on learned features and optimization (Yosinski et al., 2014; Neyshabur et al., 2020). Probe design determines which information is accessible to a specified decoder (Hewitt & Liang, 2019); CKA compares representations independently of the fitted classifier (Kornblith et al., 2019). We pair adapted-head evaluation with frozen in-task and other-task probes on common labeled supports. Rank, class geometry, and same-input drift characterize feature changes, while matched attention replay measures structural retention rather than causal token importance (Jain & Wallace, 2019). Their combination distinguishes useful specialization from linear transfer under a common evaluation protocol.

3. Setup: Selective Target Adaptation

3.1. Source Teacher and Dual Support

Definition 1 (Source teacher and target pool)**.** Let $S = \{(x_j^S, y_j^S)\}_{j=1}^{n_S}$ be a labeled source corpus and $T = \{x_i\}_{i=1}^{n_T}$ an unlabeled target pool. A source-trained encoder ϕ_{θ_0} and prediction head ω_0 define the conditional model

$$p_v(y \mid x) = \mathrm{softmax}(h_\omega(\phi_\theta(x)))_y, \quad v = (\theta, \omega). \quad (1)$$

The teacher is $v_0 = (\theta_0, \omega_0)$; only target inputs, not their labels, are available for adaptation.

The frozen teacher supplies $\hat{y}_i = \arg\max_y p_{v_0}(y \mid x_i)$ and maximum-softmax confidence $c_i = \max_y p_{v_0}(y \mid x_i)$ (Hendrycks & Gimpel, 2017).

Let $e(x)$ be a frozen semantic embedding normalized to unit length and $\mathcal{N}_k(x)$ its k nearest source neighbors. We define semantic support for the *teacher's predicted class* as

$$\pi_{ij} = \frac{\exp(e(x_i)^\top e(x_j^S)/\tau)}{\sum_{r\in\mathcal{N}_k(x_i)} \exp(e(x_i)^\top e(x_r^S)/\tau)}, \tag{2}$$

$$s_i = \sum_{j\in\mathcal{N}_k(x_i)} \pi_{ij}\mathbf{1}[y_j^S = \hat{y}_i]. \tag{3}$$

We use $k = 20$, $\tau = 0.1$, and external LaBSE embeddings (Feng et al., 2022). Source annotation or translation errors can propagate into the neighbor-support scores.

To compare signals on a common scale, let F_c, F_s be empirical target-pool percentile transforms. Dual ranks examples by the weaker percentile-transformed signal:

$$u_i = \min\{F_c(c_i), F_s(s_i)\}, \qquad w_i = \mathbf{1}[i \in \mathrm{Top}_{\rho n_T}(u)]. \tag{4}$$

We use $\rho = 0.6$ with deterministic identity-based ties. Confidence/Semantic rank by c_i/s_i at equal retained count; All retains every target input. This general-task gate and the original review–aspect cosine/absolute-confidence gate use separate protocols (Appendix D).

3.2. Attention Anchoring and Head Reset

Write $a_\theta^{\ell ht}(x)$ for the attention probability vector over valid keys at layer ℓ, head h, and query t. Padding positions and, for causal models, future positions are masked; teacher and student rows are renormalized over valid keys. Following attention-distribution distillation (Wang et al., 2020), we measure per-input teacher–student discrepancy as

$$d_A(x;\theta) = \frac{1}{|\mathcal{I}_x|} \sum_{(\ell,h,t)\in\mathcal{I}_x} \mathrm{KL}\big(\mathrm{sg}[a_{\theta_0}^{\ell ht}(x)] \| a_\theta^{\ell ht}(x)\big). \tag{5}$$

Our implementation weights valid query/head rows equally in the final layer. The adaptation objective is

$$\mathcal{L}_w(v) = \frac{\sum_i w_i[-\log p_v(\hat{y}_i \mid x_i) + \lambda d_A(x_i;\theta)]}{\sum_i w_i}. \tag{6}$$

Rejected inputs have zero direct training weight. KL acts on *attention distributions*; output-label KL defines a different objective even when its gradients reach attention parameters. Anchored variants use $\lambda = 0.1$ throughout adaptation.

Head reset redirects feature updates (Li et al., 2020). We reset at adaptation onset, whereas RIFLE resets periodically. Qwen resets its linear classification head, preserving the vocabulary head; XLM-R resets its dense–tanh–output module. Dual+KL and Dual+reinit isolate operations; Full combines both on identical Dual support.

Appendix B gives the training procedure in Algorithm 1, together with implementation and evaluation details.

4. Formal Properties of Selective Adaptation

Problem 1 (Support, structural retention, and readout risk)**.** Fix a source teacher, target pool, and adaptation budget (Definition 1). Characterize the selected update's deviation from the latent target gradient, feature changes compatible with attention retention, and the separation of adapted-head risk from best linear-readout risk.

4.1. Selection Changes Both Error and Update Support

Let $\mathbb{E}_T$ denote expectation under the latent labeled target distribution. For cross-entropy ℓ_v, define $g^* = \nabla_v \ell_v(x, y)$, $g_T = \mathbb{E}_T g^*$, and $\delta = \nabla_v \ell_v(x, \hat{y}) - g^*$. The fixed teacher and gate $w \in \{0, 1\}$ give retention $\rho = \mathbb{E}_T w > 0$ and $g_w = \mathbb{E}_T[w(g^* + \delta)]/\rho$. Finite gradient second moments suffice; this fixed-gate identity requires neither calibration nor the expansion assumption of Wei et al. (2021).

Proposition 1 (Selected-update decomposition)**.** *The expected selected pseudo-label gradient satisfies*

$$g_w - g_T = \underbrace{\frac{\mathrm{Cov}_T(w, g^*)}{\rho}}_{\textit{selection shift}} + \underbrace{\frac{\mathbb{E}_T[w\delta]}{\rho}}_{\textit{selected label error}}. \tag{7}$$

If the logit Jacobian has operator norm at most M and $\epsilon_w = \Pr(\hat{y} \neq y \mid w = 1)$, then

$$\|g_w - g_T\| \le \sqrt{\frac{1-\rho}{\rho}}\,\sigma_g + \sqrt{2}M\epsilon_w, \tag{8}$$

where $\sigma_g^2 = \mathbb{E}_T\|g^* - g_T\|^2$ is the target-gradient variance.

Appendix A proves Proposition 1 by centering wg^*, then applying Cauchy–Schwarz and Jacobian bounds.

Higher selected-label precision tightens the error bound, while support shift changes the update. Selected-label precision alone therefore does not determine alignment with the target gradient. Both terms depend on current parameters; neither score estimates the unknown target gradient.

For L-smooth $R_T(v) = \mathbb{E}_T \ell_v(x, y)$ with $\nabla R_T = g_T$, a selected update obeys the following smoothness bound

$$R_T(v - \eta g_w) - R_T(v) \le -\eta\|g_T\|^2 + \eta\|g_T\|\,\|g_w - g_T\| + \tfrac{L\eta^2}{2}\|g_w\|^2. \tag{9}$$

Corollary 1 (Conditional descent of a selected update)**.** *Let R_T be L-smooth with $L > 0$, and let κ_w denote the right-hand side of Eq. (8). If $\kappa_w < \|g_T\|$, a CE-only step strictly decreases R_T under the sufficient step-size condition*

$$0 < \eta < \frac{2\|g_T\|(\|g_T\| - \kappa_w)}{L\|g_w\|^2}. \tag{10}$$

Substitution into Eq. (9) proves the claim; $\kappa_w < \|g_T\|$ ensures a nonzero denominator.

This is a latent-gradient condition for CE-only SGD; Full's regularized optimizer is assessed empirically.

4.2. Anchoring Supplies Local Directional Curvature

Fisher geometry (Martens, 2020) yields a local metric.

Proposition 2 (Local attention metric)**.** *Use empirical averages over a fixed finite selected pool and valid rows. Fix valid-key masks and deterministic attention logits that are C^3 near θ_0, matching the teacher there. For each row, let $q = \mathrm{softmax}(b_0)$ and $J_b = \partial b/\partial\theta|_{\theta_0}$. Then*

$$R_{A,w}(\theta_0 + \Delta\theta) = \tfrac{1}{2}\Delta\theta^\top F_{A,w}\Delta\theta + O(\|\Delta\theta\|^3), \tag{11}$$

where $R_{A,w}(\theta) = \mathbb{E}_{T|w=1} d_A(x;\theta)$ and

$$F_{A,w} = \mathbb{E}_{T|w=1}\mathbb{E}_{\ell,h,t}[J_b^\top(\mathrm{diag}(q) - qq^\top)J_b]. \tag{12}$$

The matrix satisfies $F_{A,w} \succeq 0$.

Corollary 2 (Attention-compatible feature change)**.** *Let $J_\phi(x)$ be the Jacobian of a C^2 encoder at θ_0. If $d \in \ker F_{A,w}$ and $J_\phi(x)d \neq 0$ for a specified input, anchoring has zero quadratic cost while feature change has the nonzero linear term $\phi_{\theta_0+td}(x) - \phi_{\theta_0}(x) = tJ_\phi(x)d + O(t^2)$.*

Local compatibility differs from finite-displacement invariance. Appendix A gives Taylor proofs under the stated smoothness and direction conditions.

4.3. Head Reset Redirects Feature Learning

Proposition 3 (Head-mediated feature updates)**.** *For a C^2 encoder, linear head with logits Wz, $z = \phi_\theta(x)$, and fixed pseudo-label $\hat{y}$, the CE encoder gradient is*

$$\nabla_\theta \ell = J_\phi(x)^\top W^\top(p_v - e_{\hat{y}}). \tag{13}$$

A CE-only encoder SGD step of size η on this example changes the encoded feature according to

$$\Delta z = -\eta J_\phi J_\phi^\top W^\top(p_v - e_{\hat{y}}) + O(\eta^2), \tag{14}$$

The chain rule and encoder Taylor expansion prove Proposition 3 (Appendix A). Reset can change W and its residual; nonlinear heads substitute $J_h(z)$. Initial displacement and subsequent encoder learning are distinct.

Identity 1 (Representation risk and readout gap)**.** For a common linear-head family, define $R_T^*(\theta) = \inf_W R_T(\theta, W)$ and $G_T(\theta, W) = R_T(\theta, W) - R_T^*(\theta)$. Between two encoder/head pairs with finite risks,

$$\Delta R_T = \Delta R_T^* + \Delta G_T. \tag{15}$$

by subtracting the definitions at the two checkpoints.

This identity motivates separate head and probe evaluations. Finite-support Macro-F1 measures performance; Eq. (15) separates representation risk from the readout gap.

4.4. Two Drifts and Scale-Aware Geometry

We distinguish *domain discrepancy at fixed parameters* from *adaptation drift on fixed inputs*. Let $\mu_S(\theta), C_S(\theta)$ be source feature moments and $\mu_{T,w}(\theta), C_{T,w}(\theta)$ selected-target moments. Domain diagnostics are

$$\begin{aligned} D_\mu^{S,T}(\theta) &= \|\mu_S - \mu_{T,w}\|^2, \\ D_C^{S,T}(\theta) &= \|C_S - C_{T,w}\|_F. \end{aligned} \tag{16}$$

In contrast, for fixed target IDs U, adaptation diagnostics compare $Z_0 = \phi_{\theta_0}(U)$ and $Z = \phi_\theta(U)$:

$$D_\mu^{0,t} = \|\mu(Z) - \mu(Z_0)\|^2, \quad D_C^{0,t} = \frac{\|C(Z) - C(Z_0)\|_F}{\|C(Z_0)\|_F + \varepsilon}. \tag{17}$$

Covariance alignment has been used as a training objective (Sun & Saenko, 2016); here these moments diagnose domain mismatch and temporal change separately.

Using evaluation labels only after training, we measure within- and between-class scatter,

$$W_Z = \frac{1}{n}\sum_i \|z_i - \mu_{y_i}\|^2, \qquad B_Z = \frac{1}{n}\sum_k n_k\|\mu_k - \mu\|^2, \tag{18}$$

and report B_Z/W_Z, covariance effective rank, and centered linear CKA. For covariance eigenvalue proportions r_j and centered feature matrices X, Y,

$$\begin{aligned} r_{\mathrm{eff}} &= \exp\left(-\sum_j r_j \log r_j\right), \\ \mathrm{CKA}(X,Y) &= \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F\|Y^\top Y\|_F}. \end{aligned} \tag{19}$$

We repeat scatter calculations after per-example L_2 normalization to separate directional class organization from contraction due to feature scale. For nonzero centered matrices, CKA is invariant to independent orthogonal transformations and nonzero isotropic rescaling, whereas moment drifts compare feature coordinates. High CKA can therefore coexist with mean or scale drift. Together, these diagnostics distinguish preserved relational geometry from changes in feature location, magnitude, and class organization.

5. Experimental Design

Tasks and models. XLM-R-base (Conneau et al., 2020) adapts directly on XNLI inference (Conneau et al., 2018), MASSIVE intent (FitzGerald et al., 2023), AG News topics (Zhang et al., 2015), and PAWS-X paraphrase (Yang et al., 2019). Qwen3-4B-Instruct-2507 transfers French XNLI supervision to unlabeled English inputs, then receives XNLI-en/MASSIVE-ko/AG News-en/PAWS-X-en probes.

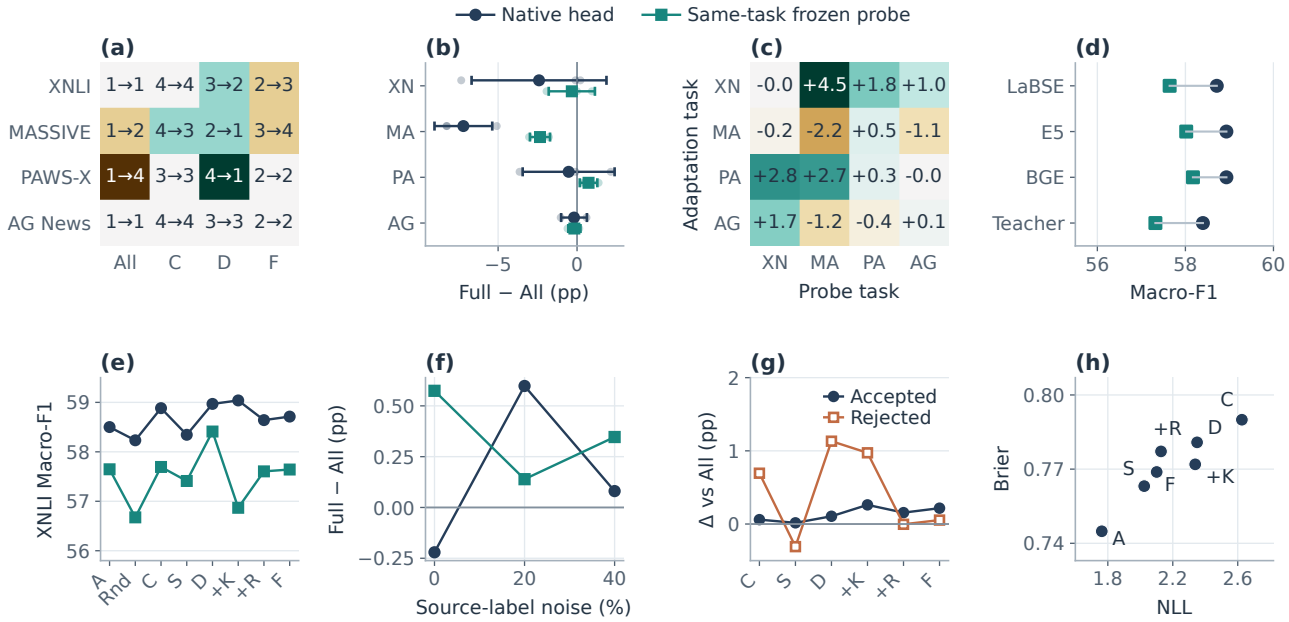


Figure 2. **XLM-R across tasks and diagnostic endpoints.** (a) Three-seed mean ranks among All, Confidence, Dual, Full for each task/readout (lower is better); color is probe minus head rank. (b) Paired Full-minus-All means ± SD. (c) Full-minus-All probe matrix. XNLI: (d) Full with alternative scorers; (e) component head/probe scores; (g) accepted/rejected test F1 minus All (3,034/1,976 inputs); (h) head NLL/Brier. (f) AG News synthetic source-label noise. Panels (a,b) aggregate source/student repeats; (c–h) show matched diagnostics. Teacher in (d) uses frozen source-encoder features to construct semantic neighborhoods.

Controlled adaptation. Qwen uses 4-bit QLoRA (Dettmers et al., 2023), rank 16, and batch 16. Selectors retain 4,800/8,000 inputs. Student seeds share teacher, gate, and source noise. Budgets are 150 and staged 600 updates (150 plus 450 with optimizer/scheduler restart, no second reset). Differentiable student attention and a frozen teacher enable KL; checkpoints are final predetermined updates.

Readouts and repetition. Frozen features receive standardized logistic-regression probes ($C = 1$) with fixed labeled supports (Appendix B). Macro-F1 means and sample SD use three seeds; cross-task scores first average non-XNLI probes within seed. Qwen repeats student optimization; XLM-R repeats source/student training. Scorer/noise sweeps use fixed checkpoints.

Mechanism measurements. Geometry uses 512 matched Qwen inputs; XLM-R replay on shared inputs reproduces saved probabilities under GPU BF16. Layers/heads are repeated measurements. ABSA retains its original protocol. In Figure 2, A/C/S/D/F/Rnd mean All, Confidence, Semantic, Dual, Full, and Random; +K/+R add KL/reset. XN/MA/PA/AG denote XNLI, MASSIVE, PAWS-X, and AG News in the general-task panels.

6. Results

Full leads the staged-600 head mean (Table 1), exceeding All by a paired 5.60 ± 4.91 points, positive in two of three seeds. At 150, Dual+KL leads at 64.54.

6.1. Full Specializes; Dual Preserves Transfer

At 600, Dual leads other-task transfer: $+1.05 \pm 0.37$ over All and $+0.46 \pm 0.16$ over Confidence, positive in every seed. Its head gain over Confidence is 2.97 ± 0.73 (Table 3).

Confidence retains 4,184 correct and 616 incorrect pseudo-labels; Dual retains 3,838 and 962 (87.17% versus 79.96% accuracy). Their rejected pools contain 1,502 and 1,848 correct pseudo-labels, respectively. Rejected pools contain both correct and incorrect pseudo-labels. Post-training gold-label counts distinguish precision from update support (Proposition 1). Equal retention (4,800) and budgets permit different identities, class balance, and semantic coverage.

6.2. The Component Effects Depend on the Endpoint

Generalization also changes outside the accepted region. On original-Dual test partitions, staged-600 Dual scores 59.74 Macro-F1 on 3,041 accepted and 1,969 rejected inputs; Confidence scores 58.57/55.04. Paired gains are $1.17 \pm 0.58 / 4.70 \pm 1.44$ points, positive in every seed. These fixed strata measure shared-parameter generalization, with zero direct weight on rejected training inputs. Class composition and semantic coverage remain candidate mediators. Figure 3 holds support fixed to compare the empirical effects of attention anchoring and head reset.

Without reset, KL yields a mean 150-step head gain of 1.59 ± 0.25 points, positive in every seed, while the other-task mean changes by 0.09 ± 0.34. At 600, differences

Table 1. **Adaptation and readout performance.** Qwen3-4B-Instruct-2507 adapted on XNLI; Macro-F1 mean ± SD across three fixed-teacher student seeds. Cross-task means average Table 2's three probes; bold/shading mark column maxima. The staged schedule restarts optimization after 150 updates. Sources: [a]Hendrycks & Gimpel (2017); [b]Feng et al. (2022); [c]Wang et al. (2020); [d]Li et al. (2020).

Adaptation variant	150 updates			600 staged updates		
	Native head	In-task probe	Cross-task mean	Native head	In-task probe	Cross-task mean
Qwen3 (Qwen Team, 2025); XNLI (Conneau et al., 2018)						
All	62.79 ± 2.75	86.38 ± 0.32	78.44 ± 0.14	60.58 ± 1.87	86.49 ± 0.33	77.99 ± 0.19
Confidence[a]	64.40 ± 3.72	86.11 ± 0.23	78.52 ± 0.33	59.20 ± 1.95	86.23 ± 0.55	78.58 ± 0.21
Semantic[b]	63.07 ± 1.89	**86.76** ± 0.33	78.44 ± 0.51	63.35 ± 1.65	**86.55** ± 0.35	78.45 ± 0.09
Dual	62.95 ± 2.07	86.52 ± 0.10	78.83 ± 0.55	62.17 ± 1.25	86.26 ± 0.48	**79.04** ± 0.28
Dual+KL[c]	**64.54** ± 1.91	86.41 ± 0.28	**78.92** ± 0.26	63.39 ± 1.13	85.90 ± 0.14	78.89 ± 0.31
Dual+reinit[d]	64.23 ± 5.83	86.68 ± 0.73	78.02 ± 0.15	65.05 ± 4.66	85.98 ± 0.57	78.12 ± 0.69
Full	64.11 ± 5.28	86.53 ± 0.22	78.14 ± 0.14	**66.19** ± 4.12	86.20 ± 1.26	77.59 ± 0.13

Table 2. **Cross-task linear-probe transfer.** Staged-600 Qwen3; Macro-F1 mean ± SD across three seeds with fixed labeled supports. Shading marks task maxima. Benchmarks: MASSIVE (FitzGerald et al., 2023), AG News (Zhang et al., 2015), and PAWS-X (Yang et al., 2019).

Adaptation variant	MASSIVE	AG News	PAWS-X
	Intent (ko)	Topic (en)	Paraphrase (en)
All	58.79 ± 0.26	88.30 ± 0.22	86.87 ± 0.21
Confidence	60.48 ± 0.12	88.25 ± 0.07	87.01 ± 0.59
Semantic	59.95 ± 0.92	88.10 ± 0.12	87.29 ± 0.78
Dual	**60.91** ± 1.18	88.47 ± 0.21	**87.73** ± 0.33
Dual+KL	60.66 ± 0.31	**88.61** ± 0.07	87.41 ± 0.72
Dual+reinit	60.40 ± 1.34	87.64 ± 0.28	86.32 ± 1.34
Full	58.88 ± 0.35	88.04 ± 0.45	85.85 ± 0.46

Table 3. **Adaptation-budget sensitivity.** Qwen3/XNLI, paired mean ± SD over three student seeds. Full's positive head and negative transfer changes occur in every seed. The 600-step continuation restarts optimizer and scheduler.

Adaptation variant	Paired Δ Macro-F1 (600−150)	
	Native head	Cross-task probe
All	−2.21 ± 4.46	−0.45 ± 0.22
Confidence	−5.20 ± 1.88	+0.06 ± 0.27
Semantic	+0.28 ± 3.28	+0.01 ± 0.58
Dual	−0.78 ± 3.11	+0.21 ± 0.48
Dual+KL	−1.15 ± 1.03	−0.03 ± 0.56
Dual+reinit	+0.82 ± 1.46	+0.10 ± 0.79
Full	+2.08 ± 1.88	−0.55 ± 0.23

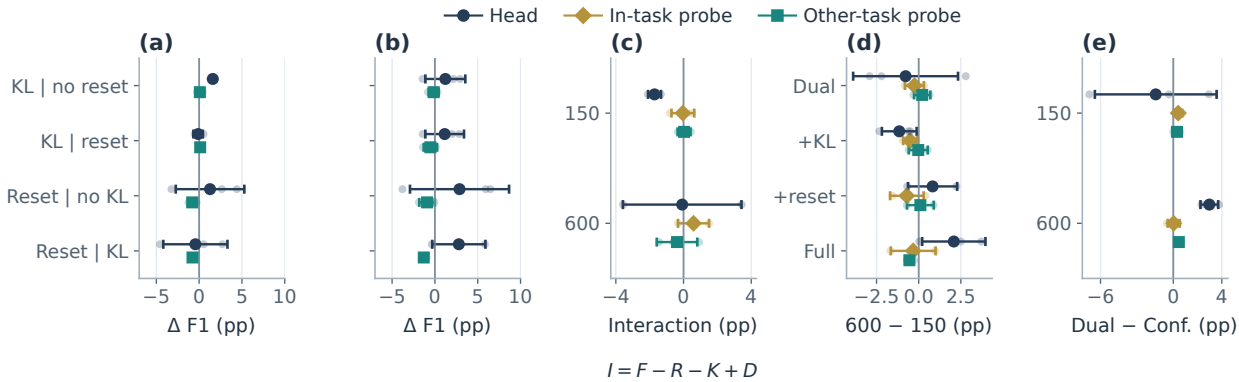


Figure 3. **Component, interaction, and budget effects.** (a,b) Factorial contrasts at 150 and staged 600, combining native-head and cross-task probe endpoints. (c) KL–reset interaction, Eq. (20); (d) paired 600-minus-150 changes across four factorial variants; (e) Dual-minus-Confidence at each budget. Panels (c–e) include native, in-task, and cross-task readouts. Markers/whiskers are paired mean ± SD over three fixed-teacher student seeds; faint points show individual differences.

are 1.22 ± 2.33 and -0.14 ± 0.58. KL's observed gains therefore depend on the evaluation endpoint.

At 600, reset increases Dual's mean head by 2.88 points and reduces other-task transfer by 0.92; on Dual+KL, changes are +2.80/−1.30. Across budgets and KL conditions, reset lowers transfer in every matched seed. Full's continuation improves its head by 2.08 ± 1.88 and reduces transfer by 0.55 ± 0.23, both directions shared by all seeds. These contrasts identify reset-driven specialization whose head benefit varies with adaptation budget.

For any endpoint m, the factorial interaction is

$$I_m = (m_{\text{Full}} - m_{\text{Dual+reinit}}) - (m_{\text{Dual+KL}} - m_{\text{Dual}}). \tag{20}$$

The head interaction is -1.71 ± 0.38 points at 150 and -0.08 ± 3.50 at 600 (paired mean ± SD). The contrast isolates how reset changes the effect of KL on a fixed endpoint. At 150 updates, its negative mean indicates attenuation of

Table 4. **Class geometry across adaptation variants.** Qwen3/XNLI, staged 600: 512 matched inputs, three seeds. Paired changes from All are mean ± SD; scatter/ratio use percentages, rank absolute differences. Shading identifies Full. Appendix F includes both budgets and temporal drifts.

Adaptation variant	Normalized geometry (%)		Δr_{eff}
	ΔW	$\Delta(B/W)$	
All	0	0	0
Confidence	$+40.23 \pm 11.99$	$+46.81 \pm 19.05$	-11.95 ± 6.56
Semantic	$+75.95 \pm 4.60$	$+13.18 \pm 8.52$	$+2.74 \pm 1.48$
Dual	$+118.65 \pm 10.68$	$+41.46 \pm 22.91$	-7.16 ± 4.29
Dual+KL	$+20.17 \pm 8.38$	$+56.81 \pm 16.50$	-14.17 ± 4.74
Dual+reinit	$+12.76 \pm 22.58$	$+169.01 \pm 31.29$	-34.45 ± 2.16
Full	-8.31 ± 7.47	$+181.74 \pm 62.53$	-36.06 ± 3.10

Table 5. **Attention structure under component ablations.** XLM-R on XNLI: matched GPU BF16 teacher–student replay. Entropy uses K valid keys and averages layers and heads.

Adaptation variant	Teacher–student JS		Entropy
	All layers	Final layer	$H/\log K$
XLM-R (Conneau et al., 2020); XNLI attention			
All	0.0400	0.1524	0.5203
Confidence	0.0587	0.2248	0.5473
Semantic	0.0466	0.1655	0.5294
Dual	0.0480	0.1720	0.5302
Dual+KL	0.0194	0.0384	0.5012
Dual+reinit	0.0561	0.2076	0.5452
Full	0.0237	0.0403	0.5075

Table 6. **Original native-review ABSA results.** XLM-R transfers translated SemEval-2014 (Pontiki et al., 2014) to KR3. ACD uses Micro-F1; polarity uses accuracy. Training tokens fall from 1.51M to 0.89M. Shading marks the highest reported point estimate separately within each training order.

Training regime	Detection	Polarity accuracy	
	ACD μF1	4-way	Binary
Joint, unfiltered	87.57	83.43	89.34
Joint, KPC-CF	87.75	83.52	89.56
Sequential, unfiltered	87.69	84.21	90.35
Sequential, KPC-CF	**88.29**	**84.34**	**90.89**

Table 7. **Original zero-shot semantic transfer.** ABSA-trained XLM-R with intermediate-layer mean pooling on KorSTS (Ham et al., 2020) and KLUE-STS (Park et al., 2021). Original-protocol correlations use no trained probe; shading marks column maxima.

Training regime	Spearman $\rho \times 100$	
	KorSTS	KLUE-STS
Pretrained	11.41	−1.23
Translated source	10.26	2.55
Target pseudo-labels	12.18	7.59
Source then unfiltered target	9.21	6.35
KPC-CF	**16.26**	**20.50**

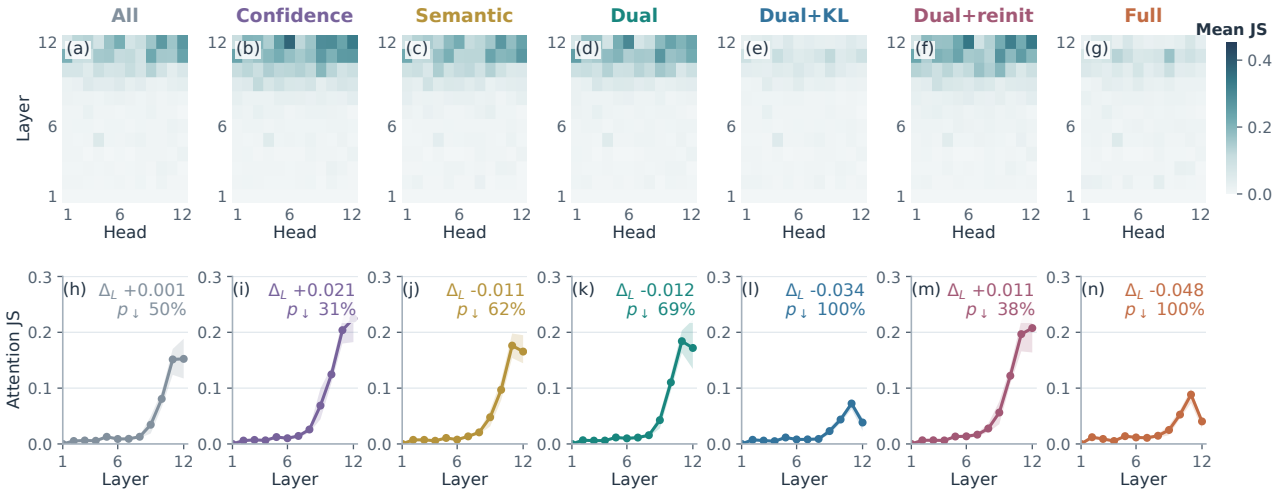


Figure 4. **Attention anchoring across layers and heads.** XLM-R/XNLI; matched GPU BF16 replay reproduces saved probabilities. Top (a–g): input-mean teacher–student JS by layer/head on a shared scale. Bottom (h–n): layer means with input interquartile ranges after head averaging. Insets report mean last-minus-penultimate JS (Δ_L) and the percentage of inputs with a final-layer decrease ($p_\downarrow$). Bands describe variation across the matched inputs within each trained checkpoint.

the head gain from KL after reset. At 600, the mean approaches zero while seed-to-seed variation increases. With support fixed, this contrast compares anchoring across readout initializations, motivating joint analysis of Full's head gains, probe responses, and representation geometry under the matched short- and long-budget conditions.

6.3. Geometry Identifies the Form of Specialization

At 150/600 updates, Full minus All changes raw B/W by 156%/193%, normalized B/W by approximately 149%/182%, and effective rank by $-27.46/-36.06$. Raw within-scatter falls 6.85%/18.76%; normalized scatter rises

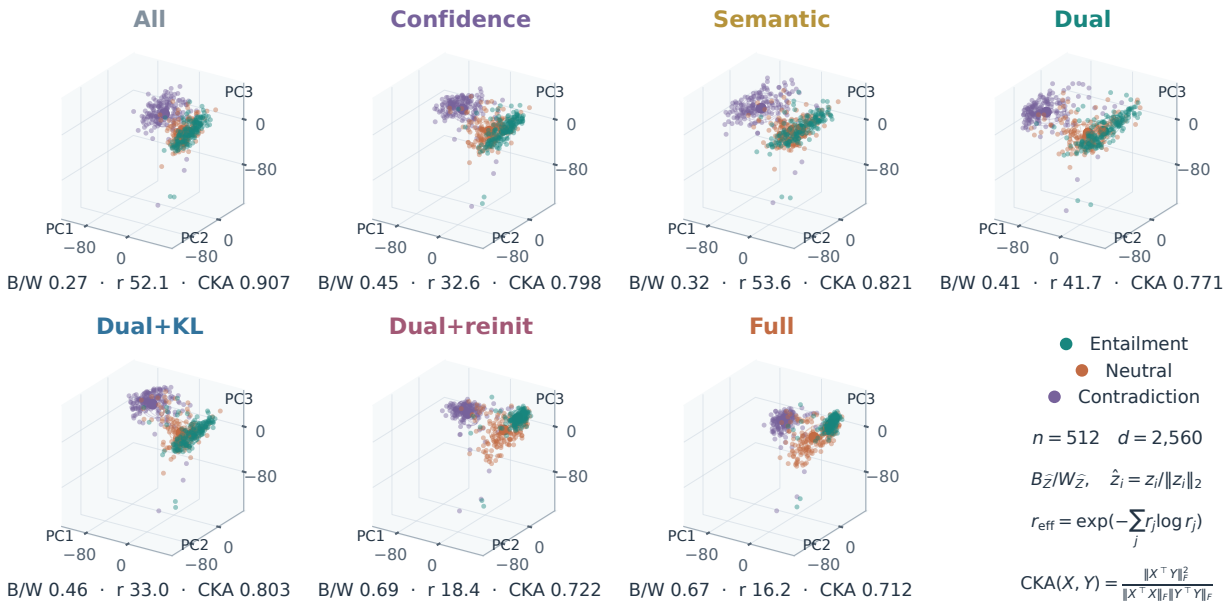


Figure 5. **Matched Qwen representation geometry.** XNLI, staged 600: seven variants share 512 inputs, pooled PCA (eight conditions; 47.03% explained), axes, and camera. Large points are centroids; wireframes are projected 1-SD covariance ellipsoids. Full-dimensional metrics (Eqs. 18–19): L_2-normalized B/W, covariance effective rank, and centered linear CKA to the source teacher.

5.51% at 150 but falls 8.31% at 600. Despite greater class separation, in-task Macro-F1 changes by -0.29 at 600 (Figure 5; Appendix F). KL changes teacher-reference CKA from 0.771/0.722 to 0.803/0.712 without/with reset. Teacher-reference drifts locate specialization in same-input reorganization (Table 13); Corollary 2 allows attention retention alongside feature change.

6.4. Attention Anchoring Acts on Its Intended Structure

Matched XLM-R replay (Figure 4) shows anchoring without/with reset (Proposition 2): KL reduces mean JS from 0.0480/0.0561 to 0.0194/0.0237 and final-layer JS from 0.1720/0.2076 to 0.0384/0.0403 (approximately 78%/81%). Teacher-relative LoRA map changes avoid factor ambiguity:

$$\Delta W_{\text{eff}} = \frac{\alpha}{r}(B_t A_t - B_0 A_0), \qquad u_W = \frac{\|\Delta W_{\text{eff}}\|_F}{\sqrt{|W|}}. \tag{21}$$

Qwen at 150 updates: aggregate effective-adapter Frobenius norms 2.16/2.66/2.67 for Dual/Dual+reinit/Full show encoder parameter updates after reset.

6.5. Operating Regimes Across Tasks and Backbones

All leads the three-seed adapted-head means on all four XLM-R tasks (Figure 2; Appendix C). Dual instead improves in-task probes over All on MASSIVE (54.79 versus 54.20) and PAWS-X (69.39 versus 68.26).

Under the original ABSA protocol (Tables 6–7), sequential XLM-R improves ACD Micro-F1/binary-polarity accuracy by 0.60/0.54 points and both STS correlations while using 0.89M versus 1.51M training tokens.

7. Discussion and Conclusion

Selective adaptation controls both supervision support and feature updates. Our selected-gradient decomposition separates label error from support shift; attention geometry and head-mediated updates explain how KL and reset constrain different parts of adaptation. These mechanisms connect the complete configuration to endpoint-specific behavior through component effects and interactions. Fixed-teacher repetitions and common probe supports isolate student adaptation under the same evaluation protocol.

In the repeated Qwen regime, Full leads the longer-budget adapted-head mean, whereas Dual leads mean cross-task linear transfer. Concentrated target geometry accompanies Full's head improvement without a corresponding fresh in-task probe gain. Reject-region gains reveal shared-parameter generalization beyond the directly supervised training support. Matched attention replay confirms structural anchoring, and the two drift definitions distinguish domain discrepancy from same-input temporal change. Original ABSA results identify a regime where Full benefits task performance and zero-shot semantic transfer; the broader XLM-R comparisons establish dependence on the task and backbone. The combined evidence supports selecting interventions by their intended readout and reporting native performance, transferable linear information, and structural retention as complementary outcomes.

Impact Statement

This study uses existing data for lower-cost adaptation. Aggregate gains establish neither fairness nor suitability for consequential decisions. Deployment evaluation should cover target performance and retained capabilities across all intended deployment populations.

References

Bao, W., Zhao, Y., Yang, X., and He, J. Ramen: Robust test-time adaptation of vision-language models with active sample selection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings*, pp. 9632–9642, 2026. URL https://openaccess.thecvf.com/content/CVPR2026F/html/Bao_Ramen_Robust_Test-Time_Adaptation_of_Vision-Language_Models_with_Active_Sample_CVPRF_2026_paper.html.

Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., and Vaughan, J. W. A theory of learning from different domains. *Machine Learning*, 79:151–175, 2010. doi: 10.1007/s10994-009-5152-4.

Chen, H., Tao, R., Fan, Y., Wang, Y., Wang, J., Schiele, B., Xie, X., Raj, B., and Savvides, M. SoftMatch: Addressing the quantity-quality trade-off in semi-supervised learning. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=ymt1zQXBDiF.

Cho, Y., Kim, Y., Yoon, J., Hong, S., and Lee, D. Feature augmentation based test-time adaptation. In *Proceedings of the Winter Conference on Applications of Computer Vision*, pp. 6838–6847, 2025. URL https://openaccess.thecvf.com/content/WACV2025/html/Cho_Feature_Augmentation_Based_Test-Time_Adaptation_WACV_2025_paper.html.

Conneau, A., Rinott, R., Lample, G., Williams, A., Bowman, S. R., Schwenk, H., and Stoyanov, V. XNLI: Evaluating cross-lingual sentence representations. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2475–2485, 2018. doi: 10.18653/v1/D18-1269. URL https://aclanthology.org/D18-1269/.

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 8440–8451, 2020. doi: 10.18653/v1/2020.acl-main.747. URL https://aclanthology.org/2020.acl-main.747/.

Dettmers, T., Pagnoni, A., Holtzman, A., and Zettlemoyer, L. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/1feb87871436031bdc0f2beaa62a049b-Abstract-Conference.html.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp. 4171–4186, 2019. URL https://aclanthology.org/N19-1423/.

Feng, F., Yang, Y., Cer, D., Arivazhagan, N., and Wang, W. Language-agnostic BERT sentence embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 878–891, 2022. doi: 10.18653/v1/2022.acl-long.62. URL https://aclanthology.org/2022.acl-long.62/.

FitzGerald, J., Hench, C., Peris, C., Mackie, S., Rottmann, K., Sanchez, A., Nash, A., Urbach, L., Kakarala, V., Singh, R., Ranganath, S., Crist, L., Britan, M., Leeuwis, W., Tur, G., and Natarajan, P. MASSIVE: A 1m-example multilingual natural language understanding dataset with 51 typologically-diverse languages. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023. URL https://aclanthology.org/2023.acl-long.235/.

Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., and Lempitsky, V. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016. URL https://jmlr.org/papers/v17/15-239.html.

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pp. 1321–1330, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

Ham, J., Choe, Y. J., Park, K., Choi, I., and Soh, H. KorNLI and KorSTS: New benchmark datasets for Korean natural language understanding. In Cohn, T., He, Y., and Liu, Y. (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 422–430, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.findings-emnlp.39. URL

https://aclanthology.org/2020.findings-emnlp.39.

Hendrycks, D. and Gimpel, K. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017. URL https://arxiv.org/abs/1610.02136.

Hewitt, J. and Liang, P. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pp. 2733–2743, 2019. doi: 10.18653/v1/D19-1275. URL https://aclanthology.org/D19-1275/.

Jain, S. and Wallace, B. C. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*, pp. 3543–3556, 2019. doi: 10.18653/v1/N19-1357. URL https://aclanthology.org/N19-1357/.

Kim, B., Ahn, C., Baddar, W. J., Kim, K., Lee, H., Ahn, S., Han, S., Suh, S., and Yang, E. Test-time ensemble via linear mode connectivity: A path to better adaptation. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/c242f2b7f60d8c685b6481939330e241-Abstract-Conference.html.

Kornblith, S., Norouzi, M., Lee, H., and Hinton, G. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 3519–3529, 2019. URL https://proceedings.mlr.press/v97/kornblith19a.html.

Kumar, A., Raghunathan, A., Jones, R., Ma, T., and Liang, P. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations*, 2022. URL https://arxiv.org/abs/2202.10054.

Lee, D.-H. Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks. In *ICML Workshop on Challenges in Representation Learning*, 2013. URL https://www.kaggle.com/blobs/download/forum-message-attachment-files/746/pseudo_label_final.pdf.

Lee, J., Jung, D., Lee, S., Park, J., Shin, J., Hwang, U., and Yoon, S. Entropy is not enough for test-time adaptation: From the perspective of disentangled factors. In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=9w3iw8wDuE.

Li, X., Xiong, H., An, H., Xu, C.-Z., and Dou, D. RIFLE: Backpropagation in depth for deep transfer learning through re-initializing the fully-connected LayEr. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 6010–6019, 2020. URL https://proceedings.mlr.press/v119/li20r.html.

Liang, J., Hu, D., and Feng, J. Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 6028–6039, 2020. URL https://proceedings.mlr.press/v119/liang20a.html.

Martens, J. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020. URL https://jmlr.org/papers/v21/17-678.html.

Neyshabur, B., Sedghi, H., and Zhang, C. What is being transferred in transfer learning? In *Advances in Neural Information Processing Systems*, volume 33, pp. 512–523, 2020. URL https://papers.nips.cc/paper/2020/hash/0607f4c705595b911a4f3e7a127b44e0-Abstract.html.

Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J. V., Lakshminarayanan, B., and Snoek, J. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html.

Park, S., Moon, J., Kim, S., Cho, W. I., Han, J. Y., Park, J., Song, C., Kim, J., Song, Y., Oh, T., Lee, J., Oh, J., Lyu, S., Jeong, Y., Lee, I., Seo, S., Lee, D., Kim, H., Lee, M., Jang, S., Do, S., Kim, S., Lim, K., Lee, J., Park, K., Shin, J., Kim, S., Park, L., Oh, A., Ha, J.-W., and Cho, K. KLUE: Korean language understanding evaluation. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021. URL https://openreview.net/forum?id=q-8h8-LZiUm.

Pontiki, M., Galanis, D., Pavlopoulos, J., Papageorgiou, H., Androutsopoulos, I., and Manandhar, S. SemEval-2014 task 4: Aspect based sentiment analysis. In Nakov, P.

and Zesch, T. (eds.), *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)*, pp. 27–35, Dublin, Ireland, August 2014. Association for Computational Linguistics. doi: 10.3115/v1/S14-2004. URL https://aclanthology.org/S14-2004.

Qwen Team. Qwen3-4B-Instruct-2507 model card. Hugging Face model repository, 2025. URL https://huggingface.co/Qwen/Qwen3-4B-Instruct-2507.

Sohn, K., Berthelot, D., Li, C.-L., Zhang, Z., Carlini, N., Cubuk, E. D., Kurakin, A., Zhang, H., and Raffel, C. FixMatch: Simplifying semi-supervised learning with consistency and confidence. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/06964dce9addb1c5cb5d6e3d9838f733-Abstract.html.

Sun, B. and Saenko, K. Deep CORAL: Correlation alignment for deep domain adaptation. In *Computer Vision – ECCV 2016 Workshops*, volume 9915 of *Lecture Notes in Computer Science*, pp. 443–450, 2016. doi: 10.1007/978-3-319-49409-8_35. URL https://link.springer.com/chapter/10.1007/978-3-319-49409-8_35.

Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., and Zhou, M. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33, pp. 5776–5788, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

Wang, Y., Chen, H., Heng, Q., Hou, W., Fan, Y., Wu, Z., Wang, J., Savvides, M., Shinozaki, T., Raj, B., Schiele, B., and Xie, X. FreeMatch: Self-adaptive thresholding for semi-supervised learning. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=PDrUPTXJI_A.

Wei, C., Shen, K., Chen, Y., and Ma, T. Theoretical analysis of self-training with deep networks on unlabeled data. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=rC8sJ4i6kaH.

Yang, Y., Zhang, Y., Tar, C., and Baldridge, J. PAWS-X: A cross-lingual adversarial dataset for paraphrase identification. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pp. 3687–3692, 2019. doi: 10.18653/v1/D19-1382. URL https://aclanthology.org/D19-1382/.

Yosinski, J., Clune, J., Bengio, Y., and Lipson, H. How transferable are features in deep neural networks? In *Advances in Neural Information Processing Systems*, volume 27, 2014. URL https://proceedings.neurips.cc/paper_files/paper/2014/hash/532a2f85b6977104bc93f8580abbb330-Abstract.html.

Zhang, B., Wang, Y., Hou, W., Wu, H., Wang, J., Okumura, M., and Shinozaki, T. FlexMatch: Boosting semi-supervised learning with curriculum pseudo labeling. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper/2021/hash/995693c15f439e3d189b06e89d145dd5-Abstract.html.

Zhang, X., Zhao, J., and LeCun, Y. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, volume 28, 2015. URL https://proceedings.neurips.cc/paper/2015/hash/250cf8b51c773f3f8dc8b4be867a9a02-Abstract.html.

Zhao, H., Tachet des Combes, R., Zhang, K., and Gordon, G. J. On learning invariant representations for domain adaptation. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 7523–7532, 2019. URL https://proceedings.mlr.press/v97/zhao19a.html.

Zheng, M., You, S., Huang, L., Wang, F., Qian, C., and Xu, C. SimMatch: Semi-supervised learning with similarity matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14471–14481, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Zheng_SimMatch_Semi-Supervised_Learning_With_Similarity_Matching_CVPR_2022_paper.html.

Zou, Y., Yu, Z., Kumar, B. V. K. V., and Wang, J. Unsupervised domain adaptation for semantic segmentation via class-balanced self-training. In *Proceedings of the European Conference on Computer Vision*, pp. 289–305, 2018. URL https://openaccess.thecvf.com/content_ECCV_2018/html/Yang_Zou_Unsupervised_Domain_Adaptation_ECCV_2018_paper.html.

A. Derivations and Interpretation

A.1. Proof of Proposition 1 and Corollary 1

For a fixed teacher and gate, expectation over $(x, y) \sim T$ gives

$$g_w = \frac{\mathbb{E}[w\nabla_v \ell_v(x, \hat{y})]}{\rho} = \frac{\mathbb{E}[wg^*]}{\rho} + \frac{\mathbb{E}[w\delta]}{\rho} \tag{22}$$

$$= \mathbb{E}[g^*] + \frac{\mathbb{E}[(w-\rho)(g^* - g_T)]}{\rho} + \frac{\mathbb{E}[w\delta]}{\rho}. \tag{23}$$

This is Eq. (7). By Cauchy–Schwarz and $w^2 = w$, the selected-gradient covariance term is bounded as

$$\left\| \frac{\mathrm{Cov}(w, g^*)}{\rho} \right\| \le \rho^{-1} \sqrt{\mathbb{E}(w-\rho)^2} \sqrt{\mathbb{E}\|g^* - g_T\|^2} = \sqrt{\frac{1-\rho}{\rho}} \sigma_g. \tag{24}$$

If $f_v(x)$ denotes the logits, the label-error contribution to the cross-entropy gradient satisfies

$$\delta = J_f(x)^\top (e_y - e_{\hat{y}}), \qquad \|\delta\| \le \sqrt{2} M \mathbf{1}[y \neq \hat{y}]. \tag{25}$$

Taking the conditional expectation on accepted inputs proves Eq. (8). The derivation applies pointwise in parameters, provided the gate is held fixed and the stated moments exist. It does not require target labels to be accessible to training. The gold-gradient quantities are analytical objects; the reported experiments do not estimate them or optimize this upper bound.

The smoothness inequality gives $R_T(v - \eta g_w) \le R_T(v) - \eta\langle g_T, g_w\rangle + L\eta^2\|g_w\|^2/2$. Substituting $g_w = g_T + (g_w - g_T)$ and applying Cauchy–Schwarz yields Eq. (9). For example, if $\|g_w - g_T\| < \|g_T\|$, sufficiently small positive η gives a descent guarantee whenever $g_w \neq 0$. The gate alone does not enforce this condition. It formalizes the distinction between reinforcing a pseudo-label and improving the true target risk under a given update.

A.2. Proof of Proposition 2 and Corollary 2

Let $A(b) = \log \sum_j \exp b_j$. For fixed teacher probabilities $q = \nabla A(b_0)$,

$$\mathrm{KL}(q \| \,\mathrm{softmax}(b_0 + \Delta b)) = A(b_0 + \Delta b) - A(b_0) - q^\top \Delta b. \tag{26}$$

A Taylor expansion of A gives the rowwise KL quadratic form, since $\nabla^2 A(b_0) = \mathrm{diag}(q) - qq^\top$. This Hessian is positive semidefinite because $u^\top(\mathrm{diag}(q) - qq^\top)u = \mathrm{Var}_{j\sim q}(u_j) \ge 0$. The all-ones vector is a null direction, reflecting softmax shift invariance. Combining the row expansion with the attention-logit Jacobian and averaging the fixed finite pool produces Eqs. (11)–(12). The expansion assumes a fixed valid-key mask and finite logits and is local around matching teacher/student logits. It does not assert a global bound on feature drift or task error. Dropout rows in the implementation are masked and renormalized; this local expansion concerns the deterministic softmax map on valid keys. For $d \in \ker F_{A,w}$, substituting $\Delta\theta = td$ removes the quadratic term, giving $R_{A,w}(\theta_0 + td) = O(|t|^3)$. Independently, the encoder Taylor expansion gives $tJ_\phi(x)d + O(t^2)$. A nonzero first-order coefficient proves Corollary 2 under its nonzero-Jacobian condition.

A.3. Proof of Proposition 3

For fixed x, cross-entropy differentiated through logits gives $\nabla_z \ell = W^\top(p_v - e_{\hat{y}})$, hence Eq. (13). Under the update $\theta' = \theta - \eta\nabla_\theta \ell$, a C^2 encoder satisfies $\phi_{\theta'}(x) = \phi_\theta(x) - \eta J_\phi \nabla_\theta \ell + O(\eta^2)$. Substitution yields Eq. (14). Both Jacobians and probabilities are evaluated before the step. For a nonlinear head, its logit Jacobian replaces W; the claim concerns a single local update and does not assert global optimizer convergence.

A.4. Identity 1 and finite probes

The identity $R_T(\theta, W) = R_T^*(\theta) + G_T(\theta, W)$ follows from the definition of G_T. Subtracting two checkpoints gives Eq. (15). The infimum avoids assuming existence of a finite optimum for separable data. With regularization, the same algebra holds if the objective and head family are defined consistently. In the experiments, a standardized, regularized, finite-support logistic regression is a controlled decoder, not an oracle achieving R_T^*. Its utility is comparability: every representation receives the same support, decoder family, and regularization. Probe labels are additional supervision for evaluation and are not part of the unsupervised target-adaptation claim.

Algorithm 1 KPC-CF: selective target adaptation with a frozen source teacher

Require: Source teacher v_0, source S, unlabeled target T, scorer e, retention ρ, KL weight λ, reset flag r.
1: Cache teacher probabilities and source/target embeddings.
2: Compute source-neighbor support using Eq. (3).
3: Fix $\hat{y}$, empirical ranks, and gate w via Eq. (4).
4: Initialize $\theta \leftarrow \theta_0$; set $\omega \leftarrow \omega_0$ or reset ω if $r = 1$.
5: **for** each predetermined adaptation update **do**
6: Sample accepted target inputs; evaluate frozen-teacher attention if $\lambda > 0$.
7: Update trainable encoder parameters and task head using Eq. (6).
8: **end for**
9: Evaluate the adapted head, then fit fresh probes on fixed labeled support with the encoder frozen.

A.5. Why the two drifts cannot be interchanged

For fixed θ, $D^{S,T}$ compares two input distributions through the same encoder. For fixed input IDs, $D^{0,t}$ compares two encoders on a fixed input distribution. A representation can preserve its target mean before/after adaptation while source–target discrepancy changes, or vice versa. Further, small differences in first and second moments do not imply pointwise feature equality, matching class conditionals, or proximity of parameter vectors. Our figures use the indicated diagnostic and preserve its conditioning variables. Reduced attention discrepancy likewise concerns observed probability distributions, rather than parameter-space proximity or preservation of individual feature vectors.

B. Protocol and Reproducibility Inventory

Table 8. **Training and evaluation protocol.** Qwen3-4B-Instruct-2507 and XLM-R-base. Update caps interact with epoch budgets and selected-pool size. The Qwen continuation restarts optimizer and scheduler at step 150. Component source superscripts follow Table 1.

Protocol factor	Model and experiment family	
	Qwen3: factorial/budget	XLM-R: general-task adaptation
Backbone	Qwen3-4B-Instruct-2507 (Qwen Team, 2025)	XLM-RoBERTa-base (Conneau et al., 2020)
Adaptation	4-bit QLoRA plus task head	Encoder plus task head
Maximum length	128	192
Effective batch size	16	32
Learning rate	10^{-4}	2×10^{-5}
Weight decay	0.01	0.01
Source teacher update cap	200	500
Student budget	150, then 450 additional	400-update cap, at most 2 epochs
Main seeds	17, 23, 42 (student)	17, 23, 41 (source/student)
Teacher/filter during repeats	Fixed to seed 17	Recomputed per source/student seed
Main retention	60%	60%
LaBSE scorer[b]	LaBSE	LaBSE
Neighborhood	20 neighbors, temperature 0.1	20 neighbors, temperature 0.1
Attention KL[c]	Teacher-to-student, final layer	Teacher-to-student, final layer
KL coefficient	0.1	0.1
Head reset[d]	Task classification head only	Task classification head only
Checkpoint selection	Final predetermined update	Final predetermined update

Qwen uses LoRA rank 16 and alpha 32. Its model revision is `cdbee75f17c01a7cc42f958dc650907174af0554`. Teacher labels and selection files are cached before adaptation. The source teacher's seed is not resampled in the three-student-seed comparison; neither are the probe supports or test sets. Student seed repetitions therefore measure conditional training variation. Qwen and XLM-R use distinct repetition schemes, and these estimates are not pooled.

For Qwen, the 150-step budget processes 2,400 sample exposures and the total 600-step budget processes 9,600 at effective batch size 16. A retained pool of 4,800 is a candidate set, not a claim that every example is visited during the first 150 updates. All uses 8,000 candidates at the same update budget. Confidence, Semantic, and Dual have identical retained counts but can differ in class composition and actual examples encountered. A same-count comparison isolates the selection rule from nominal retention, not from all downstream distributional consequences.

The frozen probes use the last valid token for Qwen and the stored encoder feature for XLM-R. Standardization is fit on

the labeled support only, followed by logistic regression at $C = 1$. The MASSIVE protocol evaluates Macro-F1 over the configured 60-class label inventory, including classes absent from a particular test subset. This convention is held fixed across methods. Cross-task means weight MASSIVE, AG News, and PAWS-X equally; task scores and student seeds are not treated as independent replicates of a single random quantity.

Fixed labeled supports contain 852/667/2,000/663 examples for XNLI/MASSIVE/AG News/PAWS-X, respectively.

KL gradient verification. The Qwen trainer uses non-reentrant checkpointing so that returned attention tensors remain connected to the adaptation graph. The revised trainer produces finite, nonzero KL-only LoRA gradients. Earlier checkpoints produced under a detached-attention implementation are not used as KL-enabled results. Dual and Dual+reinit checkpoints were reused after a two-step no-KL equivalence check of logits, gradients, and optimizer updates; this check verifies the unchanged no-KL path without claiming that a newly rerun full trajectory was compared bitwise. Gradient connectivity establishes that the regularizer is active, not that its contribution is beneficial at every endpoint.

Data and noise axes. XNLI uses a French translated source and English target. MASSIVE transfers from English to Korean; PAWS-X transfers from Korean to English. Parallel groups are assigned to source or target rather than placing both translations of a training group on opposite sides. AG News uses disjoint native-English subsets for source and target in the same topic domain. Its noise sweep randomly corrupts source labels at 0, 20, or 40%; it is not a translation-noise severity experiment or an out-of-domain benchmark. Alternative source-neighborhood encoders include LaBSE, multilingual E5, BGE, and frozen teacher features in the matched checkpoint comparison. These scorer comparisons assess sensitivity to the semantic space and do not constitute three-seed replications for every scorer.

Scope of evidence. The experiment artifacts record task datasets, partitions, source labels, student seeds, and checkpoint identities. Geometry, attention, and parameter summaries retain their own measurement scope. The repeated Qwen study supplies quantitative component comparisons. Scorer/noise sweeps and matched geometry, attention, and parameter diagnostics compare matched checkpoints. Original ABSA measurements follow their distinct application protocol; their reported estimates are interpreted separately from the new experiment's seed uncertainty.

C. Complete XLM-R General-Task Results

Table 9. **Complete XLM-R general-task evaluation.** Macro-F1 mean ± SD across three source/student seeds. Each task receives direct adaptation and a fresh probe on fixed labeled support. Shading marks row maxima, including the teacher.

Dataset / task	Readout	Source	Target adaptation variant			
		Teacher	All	Confidence	Dual	Full
AG News (Zhang et al., 2015)	Head	89.93 ± 0.31	89.85 ± 0.61	88.79 ± 0.35	89.11 ± 0.31	89.65 ± 0.28
Topic (en)	Probe	88.34 ± 0.47	89.04 ± 0.23	88.50 ± 0.12	88.78 ± 0.18	88.87 ± 0.19
MASSIVE (FitzGerald et al., 2023)	Head	28.79 ± 2.77	25.95 ± 2.26	18.65 ± 1.98	24.83 ± 2.55	18.75 ± 0.46
Intent (ko)	Probe	53.42 ± 0.09	54.20 ± 1.00	52.46 ± 0.22	54.79 ± 1.42	51.85 ± 0.53
PAWS-X (Yang et al., 2019)	Head	63.49 ± 15.24	63.36 ± 15.89	62.64 ± 11.13	61.84 ± 16.40	62.83 ± 14.18
Paraphrase (en)	Probe	66.89 ± 14.69	68.26 ± 15.57	68.39 ± 15.72	69.39 ± 17.28	68.99 ± 15.99
XNLI (Conneau et al., 2018)	Head	53.77 ± 9.28	55.87 ± 8.60	52.70 ± 13.18	53.28 ± 12.74	53.46 ± 12.75
Inference (en)	Probe	54.98 ± 1.50	57.80 ± 1.81	56.33 ± 3.49	57.56 ± 3.01	57.47 ± 3.24

Rank transitions separate classifier behavior from reusable features; proper losses quantify native-head probability quality. The large seed spread on PAWS-X and XNLI reflects the measured regime. Reporting teacher, student, and probe means together exposes inherited source-training variation. Qwen instead fixes the teacher to isolate student optimization and component effects. These repetition designs separate upstream variability from conditional adaptation effects.

D. Aspect-Specific Instantiation and Original Application Results

The original application casts review–aspect pairs as classification inputs. Its source is translated SemEval-2014 supervision (Pontiki et al., 2014) (Kor-SemEval), and its target is the original native Korean review corpus (KR3). Labels combine aspect absence with sentiment categories, enabling aspect category detection (ACD) and aspect category polarity (ACP) evaluation. The aspect gate combines review–aspect LaBSE cosine with maximum teacher probability above 0.5. The original main-table protocol reports a cosine cutoff of 0.15; the original ablation table denotes a mean-based cosine cutoff.

These reported protocols are retained separately from the rank-normalized source-neighbor gate of Eq. (4).

Table 10. **Original KR3 application results by backbone and training order.** mBERT and XLM-R transfer translated SemEval-2014 supervision (Pontiki et al., 2014) to native KR3 reviews. Original-protocol ACD Micro-F1 and polarity accuracy; shading marks maxima within each backbone/order pair of reported target-supervision variants.

Backbone	Training order	Target supervision	Detection	Polarity accuracy		
			ACD μF1	4-way	3-way	Binary
mBERT (Devlin et al., 2019)	Joint	Unfiltered	84.64	80.45	82.71	86.57
		Filtered	85.42	80.88	83.15	86.96
	Sequential	Unfiltered	83.73	78.37	80.58	84.60
		Filtered	83.47	78.94	81.16	85.09
XLM-R (Conneau et al., 2020)	Joint	Unfiltered	87.57	83.43	85.78	89.34
		KPC-CF	87.75	83.52	85.90	89.56
	Sequential	Unfiltered	87.69	84.21	86.57	90.35
		KPC-CF	88.29	84.34	86.72	90.89

The original comparisons distinguish training order from backbone capacity. Joint mBERT filtering improves ACD by 0.78 points and binary polarity by 0.39 points. Sequential XLM-R Full improves the same endpoints by 0.60 and 0.54 points. Sequential mBERT, however, improves polarity while its ACD decreases by 0.26 points. Detection and polarity therefore remain separate endpoints even within the original application.

Table 11. **Original XLM-R ablations: performance, attention change, and cost.** KR3 point estimates under the original training schedules. Attention JS compares pre- and post-adaptation attention; processed examples count both stages, and GPU time is in minutes. Component superscripts follow Table 1. Shading marks predictive maxima and the lowest reported GPU time.

Adaptation variant	Detection	Polarity accuracy			Structure	Training cost	
	ACD μF1	4-way	3-way	Binary	Attn. JS	Examples	GPU min
Unfiltered reference	87.57	83.43	85.78	89.34	0.0736	30.45K	41
Semantic threshold[b]	88.44	83.31	85.66	89.78	0.0409	22.00K	31
Confidence threshold[a]	87.14	84.50	86.88	90.86	0.0681	29.53K	38
Dual + KL (original)[c]	87.75	83.52	85.90	89.56	0.0457	21.30K	30
Dual + reinitialization[d]	87.69	84.21	86.57	90.35	0.0598	21.30K	30
Full hard gate	88.29	84.34	86.72	90.89	0.0438	21.30K	29
Full soft gate	87.95	72.47	74.51	78.09	0.0506	21.30K	32

Table 11 connects component effects to structural change and computational cost. Semantic selection has the strongest ACD result, while Full combines the highest binary polarity score with 21.30K processed examples and 29 GPU minutes. Across the hard-gate component variants, Full pairs lower attention JS than either partial variant with higher detection and polarity scores. These original operating points complement the matched-budget Qwen contrasts by relating target prediction to attention retention and the amount of processed target supervision.

Table 12. **Shallow-feature readouts and predictive reliability.** (a) Logistic regression on neural token embeddings or TF-IDF: filtered minus unfiltered target supervision in combined source–target data, in percentage points. Model references follow Table 10. (b) Five-class accuracy and probability-quality diagnostics on the same KR3 gold set; accuracy and ECE variants are percentages. Shading marks column optima, maximizing accuracy and readout gains and minimizing probability-quality diagnostics.

(a) Feature-based readout gains

Feature source	Detection Δ μF1	Polarity Δ 4-way
XLM-R token embs.	+16.01	+6.29
mBERT token embs.	+13.14	+5.19
TF-IDF token features	+0.19	+0.19

(b) Predictive accuracy and probability quality

Target training	Acc.	ECE	Adaptive ECE	NLL	Brier	None ECE
KR3 only	89.39	9.64	9.63	0.609	0.199	7.10
Unfiltered	90.19	9.15	9.06	0.660	0.188	6.16
KPC-CF	90.74	8.77	8.67	0.628	0.179	5.86

Filtering improves neural-feature readouts as well as the lexical reference (Table 12a). Full improves five-class accuracy by 0.55 points over unfiltered adaptation while lowering ECE by 0.38 points, NLL by 0.032, and Brier by 0.009. These complementary endpoints connect selective supervision to feature-based classification and predictive reliability, extending the semantic-transfer evidence in Table 7 with complementary measurements on native target reviews.

E. Measurement Details and Interpretation Boundaries

E.1. Inference and intent geometry

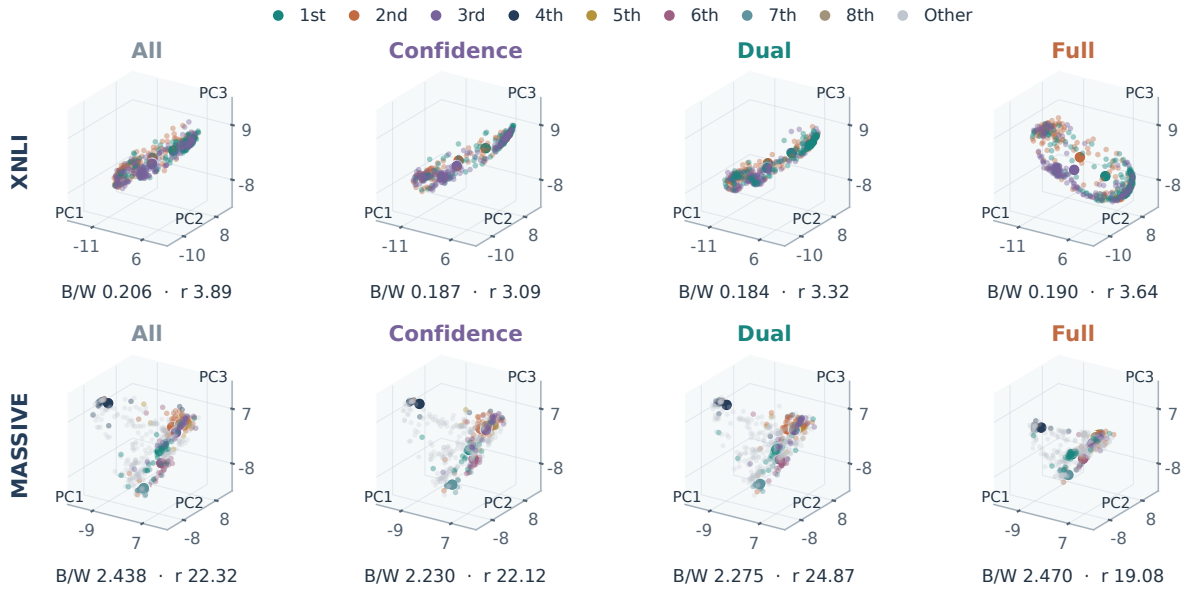


Figure 6. **Matched XLM-R geometry for inference and intent.** Rows show XNLI and MASSIVE; both use All, Confidence, Dual, and Full in aligned columns. Each task uses 512 matched inputs and its saved pooled raw-feature PCA, with shared axes and camera within the row. The eight most frequent classes are colored; remaining classes are gray and large points mark centroids. Insets give full-dimensional raw B/W and effective rank r. Displayed components explain 71.78%/30.28% of pooled variance.

The same-backbone comparison exposes task-dependent geometry. On XNLI, raw B/W is 0.206, 0.184, and 0.190 for All, Dual, and Full, respectively; effective ranks are 3.89, 3.32, and 3.64. Full's raw B/W therefore remains below All here, unlike the Qwen specialization regime, locating geometry within its backbone and reference condition. On MASSIVE, Full has a larger raw B/W than Dual (2.470 versus 2.275) and a lower effective rank (19.08 versus 24.87). Despite this class organization, the three-seed in-task probe favors Dual over Full.

MASSIVE's 512-example atlas contains 57 of 60 configured classes. Its eight most frequent classes account for 208 examples; 304 points are gray. Scatter and rank use all 512 inputs, while centroids represent only observed classes.

Geometry. Covariance effective rank is $\exp(-\sum_j r_j \log r_j)$, where r_j are nonnegative covariance eigenvalues normalized by their sum. For centered feature matrices X, Y, linear CKA is $\|X^\top Y\|_F^2/(\|X^\top X\|_F \|Y^\top Y\|_F)$. Geometry uses raw and separately L_2-normalized features, distinct from probe standardization. Labels enter only post-training scatter measurement. Each comparison shares a fixed projection, expressing every displayed condition in a common coordinate system.

Attention. Replay excludes padding and uses the valid causal mask when applicable. The descriptive XLM-R replay covers 12 layers and 12 heads on shared, hash-selected test inputs. Saved and replayed probabilities agree exactly under the matched batch/precision setting. Distributions summarize these same inputs repeatedly across layer/head structure; the resulting rows are not independent examples for significance testing. Attention Jensen–Shannon divergence is a symmetric diagnostic, whereas the optimized KL is directional teacher-to-student. Their roles and numerical scales differ.

E.2. Paraphrase and topic geometry

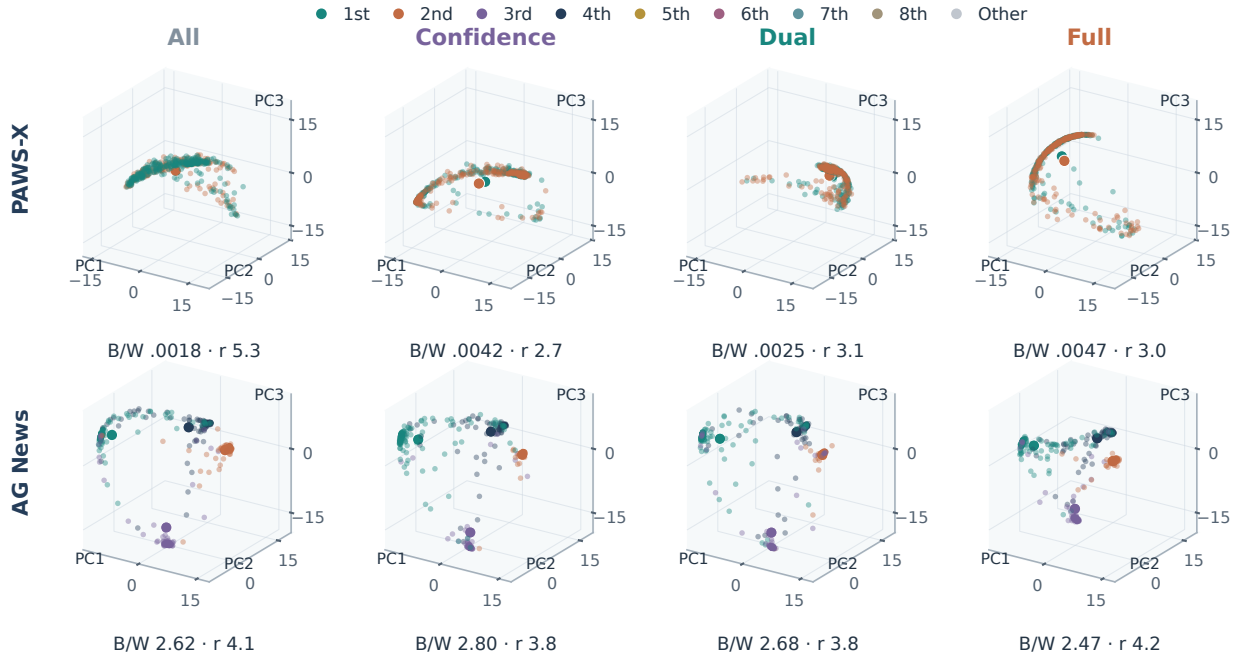


Figure 7. **Matched XLM-R geometry for paraphrase and topic classification.** Rows show PAWS-X and AG News; columns show All, Confidence, Dual, and Full. Each row contains the same 512 target inputs and a pooled PCA across saved conditions, with shared axes and camera. Colors identify classes by their within-task frequency rank; large points are class centroids. Insets show raw full-dimensional B/W and effective rank r. The displayed components explain 87.71% and 72.46% of pooled variance. AG News uses 20% synthetic source-label noise; PAWS-X uses the zero-added-label-noise condition.

PAWS-X illustrates why the scale of a ratio matters alongside its relative change. Raw B/W increases from 0.00179 for All to 0.00248 for Dual and 0.00467 for Full, while effective rank decreases from 5.27 to 3.14 and 3.04. Although the relative change is large, between-class scatter remains small compared with within-class scatter. Reporting both the absolute values and the common-coordinate projection prevents a relative percentage from implying complete separation. Both binary classes are represented, and every displayed input has an observed class label.

AG News exhibits a different ordering on the same backbone. Raw B/W is largest for Confidence (2.796), followed by Dual (2.683), All (2.617), and Full (2.470). Full instead has the largest effective rank: 4.23 versus 3.80–4.07 for the other configurations. The four topic classes all appear in the fixed subset. These measurements support a task-conditional interpretation of rank and scatter: neither lower rank nor larger B/W serves as a universal surrogate for the adapted head or a fresh probe. Matched projections locate these differences; full-dimensional statistics and distinct adapted-head and fresh-probe scores determine which geometric changes accompany useful prediction.

Parameters. Effective Qwen adapter changes incorporate LoRA scaling, so their norms describe linear-map updates rather than arbitrary factor rescalings. The analysis includes 14 checkpoints across seven configurations and two budgets. Head reset creates initial displacement by construction; subsequent backbone updates and readout/probe differences supply the informative comparison. Parameter norms characterize structural displacement and complement performance readouts.

Study boundaries. The repeated Qwen comparisons condition on a fixed teacher and filter; other-task probes measure transfer to specified linear readouts. Four directly adapted XLM-R tasks establish a different ranking. Section 4 provides identities and local expansions, rather than a universal risk guarantee. Three student seeds support paired descriptions, with model/task dependence retained in the endpoint-aware interpretation.

F. Complete Geometry and Structural Measurements

Table 13 expands the main geometry comparison to every selection and component variant at both budgets. Each contrast is calculated within seed before averaging. Zero contrasts for All define the reference; they do not indicate zero variability in its absolute geometry. All conditions use the same inputs and feature interface.

Table 13. **Class geometry and temporal drift across adaptation variants.** Qwen3/XNLI, 512 matched inputs and three fixed-teacher student seeds. Entries are paired mean ± sample SD of within-seed changes relative to All; scatter, ratio, and drift changes are percentages, while rank changes are absolute. All is zero by construction. Raw and per-input L_2-normalized geometry remain separate; teacher-relative drifts compare source and adapted encoders on identical target inputs.

Adaptation variant	Within-seed relative change (%)			Δr_{eff}	Temporal drift change (%)	
	Raw W	Normalized W	Normalized B/W		$D_\mu^{0,t}$	$D_C^{0,t}$
150 updates						
All	0	0	0	0	0	0
Confidence	+44.55±33.53	+41.81±23.82	+47.77±21.37	−7.16±9.67	+293.64±184.15	+154.48±40.92
Semantic	+5.28±9.96	+3.42±7.61	+3.58±13.98	−0.48±5.77	+38.72±25.43	+12.70±23.78
Dual	+62.01±30.16	+69.18±32.45	+30.97±14.42	−2.94±4.32	+259.85±150.00	+145.48±33.97
Dual+KL	+36.45±17.63	+38.01±14.23	+44.91±11.73	−7.49±3.16	+113.53±54.06	+123.87±19.14
Dual+reinit	−0.14±22.20	+13.33±7.85	+150.54±56.79	−27.01±6.30	+469.01±253.55	+213.92±44.91
Full	−6.85±16.98	+5.51±3.85	+148.91±53.70	−27.46±6.16	+418.43±261.75	+194.08±41.57
600 staged updates						
All	0	0	0	0	0	0
Confidence	+36.14±8.47	+40.23±11.99	+46.81±19.05	−11.95±6.56	+127.46±43.40	+123.70±25.50
Semantic	+67.38±5.49	+75.95±4.60	+13.18±8.52	+2.74±1.48	+198.64±29.89	+110.48±23.64
Dual	+90.40±8.15	+118.65±10.68	+41.46±22.91	−7.16±4.29	+384.79±69.38	+212.48±66.25
Dual+KL	+25.36±9.77	+20.17±8.38	+56.81±16.50	−14.17±4.74	+85.86±34.44	+115.63±41.15
Dual+reinit	−0.02±13.43	+12.76±22.58	+169.01±31.29	−34.45±2.16	+241.88±44.75	+218.79±73.54
Full	−18.76±3.28	−8.31±7.47	+181.74±62.53	−36.06±3.10	+118.23±51.63	+167.18±40.00

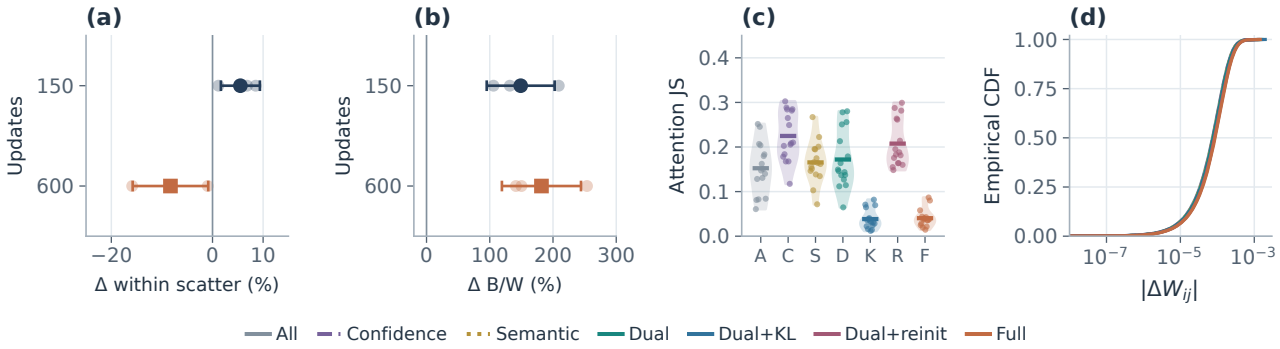


Figure 8. **Geometry and parameter updates complement attention.** (a,b) Normalized within-scatter and B/W, Full minus All (Qwen, three-seed mean ± SD); (c) seven-condition final-layer JS per input, averaging 12 heads over matched GPU BF16 XLM-R inputs; (d) seven-condition effective LoRA update ECDF (Qwen, staged 600). Ticks A/C/S/D/K/R/F follow legend order. Each panel retains its stated backbone, budget, and replication unit to distinguish these measurement scopes.

Selection and reset affect distinct summaries. At staged 600, Semantic increases effective rank relative to All, whereas reset variants reduce it strongly. Dual increases normalized within-class scatter while retaining the strongest mean cross-task probe score. Full combines reduced normalized within-class scatter with increased between/within ratio and the highest mean native-head score. These contrasts distinguish task-focused organization from broader linear decodability; they do not make effective rank or class scatter a universal performance surrogate.

G. Structural Interpretation and Layer-Resolved Attention

Relative to All, Full's raw B/W rises by 156%/193% and effective rank falls by 27.46/36.06 at 150/600 updates. Scale normalization changes the interpretation: at 150, raw within-scatter falls 6.85% but normalized scatter rises 5.51%; at 600, both fall, by 18.76%/8.31%. Normalized B/W still rises approximately 149%/182%.

Normalized within-scatter rises at 150 updates but falls at 600, whereas normalized B/W increases at both budgets. Angular class organization therefore differs from uniform contraction. Effective updates incorporate LoRA scaling and describe trained linear maps; sampled coordinates are descriptive measurements, not independent training repetitions.

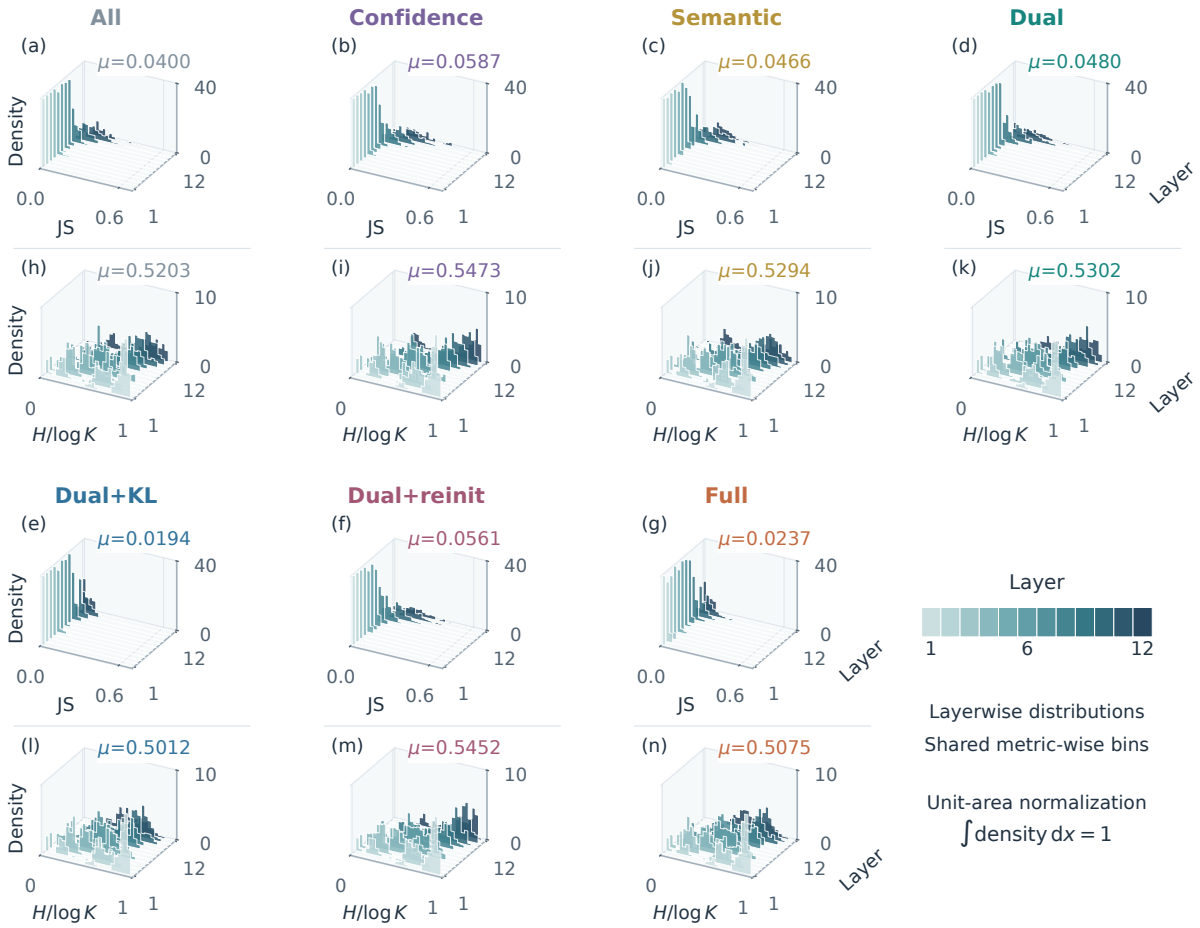


Figure 9. **Layer-resolved attention distributions.** The upper group compares All, Confidence, Semantic, and Dual; the lower group compares Dual+KL, Dual+reinit, and Full. Each condition pairs teacher–student JS (a–g) with normalized entropy (h–n). Common bins and scales compare matched XLM-R XNLI inputs and 12 heads across 12 layers using GPU BF16 replay. Annotated means average layers, heads, and inputs. Layerwise densities pool repeated input–head measurements and integrate to one. The shared reading key identifies layers and measurement units, not independent training replications.

Across selection-only variants, teacher-relative attention follows a distinct ordering: All has lower mean JS than Confidence, Semantic, and Dual. KL lowers both JS and normalized entropy in the matched no-reset and reset comparisons. Reference proximity and attention concentration characterize distinct structural properties, reported alongside separate measurements of task performance and transfer. JS is symmetric, whereas optimized KL is directional. Exact saved-probability reproduction checks the replay protocol, while the head/input distributions describe variation within a single trained seed.